

Chapter

Study Of Acids, Bases And Salts

1. A compound P is heated in a test tube with sodium hydroxide solution. A red litmus paper held at the mouth of the test tube turns blue. Which of the following could compound P be? SQP 2025

- (a) zinc sulphate
- (b) copper sulphate
- (c) ferrous sulphate
- (d) ammonium sulphate

2. Which of the following about oxides is correct? SQP 2025

- (a) A basic oxide is an oxide of a non-metal
- (b) Acidic oxides contain ionic bonds
- (c) Amphoteric oxides contain a metal
- (d) Basic oxides are always gases

3. The metal hydroxide which reacts with both acids and alkalis to form salt and water is: MAIN 2024

- (a) Calcium hydroxide
- (b) Magnesium hydroxide
- (c) Aluminium hydroxide
- (d) Ferric hydroxide

4. On passing ammonia gas over heated copper oxide for some time, a reddish-brown residue is left behind. What property of ammonia is demonstrated here? MAIN 2024

- (a) Basic property
- (b) Oxidising property

- (c) Reducing property
- (d) Acidic property

5. If the pH of a solution is '2', then the solution is a COMP 2021

- (a) strong acid
- (b) strong alkali
- (c) weak acid
- (d) weak alkali

6. The acidity of aluminium hydroxide is SQP 2021

- (a) 3
- (b) 1
- (c) 4
- (d) 2

7. Which of the following is a tribasic acid? MAIN 2021

- (a) H_2SO_4
- (b) $\text{Al}(\text{OH})_3$
- (c) H_3PO_4
- (d) $\text{Ca}(\text{OH})_2$

8. The products formed when an acid reacts with a base is COMP 2021

- (a) salt and hydrogen
- (b) salt and oxygen
- (c) salt and water
- (d) salt and carbon dioxide

9. Hydracids are those acids which contain MAIN 2021

- (a) hydrogen with any metal
- (b) hydrogen, a non-metal and oxygen
- (c) hydrogen and a non-metal other than oxygen
- (d) hydrogen and oxygen only

10. When copper oxide and dilute hydrochloric acid react, colour changes to:

MAIN 2023

- (a) white
- (b) black
- (c) blue-black
- (d) bluish-green

11. Which of the given is a strong base?

SQP 2022

- (a) Potassium hydroxide
- (b) Magnesium hydroxide
- (c) Ammonium hydroxide
- (d) Calcium hydroxide

12. Why zinc chloride is stored in air-tight bottles ?

COMP 2022

- (a) Because it is highly basic substance.
- (b) Because it is highly reactive substance.
- (c) Because it is highly acidic substance.
- (d) Because it is a deliquescent substance.

13. Which gas is evolved when HNO_3 react with sodium bicarbonate?

COMP 2021

- (a) CO_2
- (b) SO_2
- (c) H_2
- (d) H_2S

14. Which of the following statement is correct?

COMP 2023

- (a) Bases are sour in taste.
- (b) Anion can be used as an oil factory indicator to check acid or base.
- (c) Bases turn blue litmus solution to red.
- (d) pOH is defined as the logarithm of $[\text{OH}^-]$ ions.

15. Any one hydric acid

16. A strong but non-volatile mineral acid.

17. [sulphuric acid/hydrochloric acid] does not form an acid salt.

SQP 2025

18. A [reddish brown/dirty green] coloured precipitate is formed when ammonium hydroxide is added to a solution of ferrous chloride.

SQP 2025

19. An [alkaline/acidic] solution will turn methyl orange solution pink or red. SQP 2025

20. A salt formed by the complete neutralization of an acid by a base.

SQP 2025

21. From the list given below, select the word(s) required to correctly complete blanks (i) to (v) in the following passage : MAIN 2023

ammonia, ammonium carbonate, carbon dioxide, hydrogen, hydronium, hydroxide, precipitate, salt, water. A solution X turns blue litmus red, so it must contain: (i) ____ ions. Another solution Y turns red litmus blue and therefore, must contain (ii) ____ ions. When solutions X and Y are mixed together, the products will be a (iii) ____ and (iv) _____. If a piece of magnesium was put into solution X, (v) ____ gas would be evolved.

(Note : Words chosen from the list are to be used only once. Write the answers as (i), (ii), (iii) and so on. Do not copy the passage.)

22. Rewrite the following statements by adding the correct word as shown in the example: MAIN 2024

Example: Given Statement: Ammonia changes moist red litmus to blue.

Correct statement: Aqueous ammonia changes moist red litmus to blue.

(a) Sulphuric acid acts as a dehydrating agent.

(b) Ammonia reacts with chlorine to give ammonium chloride and nitrogen.

23. Hydrogen chloride gas is prepared in the laboratory by the action of concentrated sulphuric acid on sodium chloride. MAIN 2024

- (a) Give balanced chemical equation for the above reaction.
- (b) State the method of collection of the gas formed above.
- (c) What is property of sulphuric acid that makes it a suitable reagent for the reaction.

24. Name a substance which changes the blue colour of copper sulphate crystals to white. COMP 2023

Solution

1. (d) ammonium sulphate

When ammonium sulphate reacts with NaOH, **ammonia gas** is released, which is basic and turns red litmus **blue**.

2. (c) Amphoteric oxides contain a metal

Explanation: Amphoteric oxides are **metal oxides** that can react with both acids and bases.

3. (c) Aluminium hydroxide

Aluminium hydroxide is **amphoteric**, meaning it reacts with both **acids and bases** to form salts and water.

4. (c) Reducing property

Ammonia (NH_3) reduces heated **copper oxide** (CuO) to **copper** (Cu), leaving a reddish-brown residue. This demonstrates the **reducing property** of ammonia.

5. (a) Strong acid

A solution with **pH 2** has a high concentration of hydrogen ions (H^+), indicating it is strongly acidic. Lower pH values correspond to stronger acids.

6. (a) 3

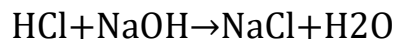
Aluminium hydroxide, $\text{Al}(\text{OH})_3$, requires **3 acidic H^+ ions** to completely neutralize it when reacting with an acid, so its acidity is 3.

7. (c) H_3PO_4

Phosphoric acid (H_3PO_4) is a **tribasic acid** because it has **three replaceable hydrogen ions (H^+)** that can react with a base in a neutralization reaction.

8. (c) salt and water

When an **acid reacts with a base**, a **neutralization reaction** occurs, producing a **salt** and **water**. For example:



9. (c) hydrogen and a non-metal other than oxygen

Hydracids are acids composed of **hydrogen and a non-metal element**, excluding oxygen. For example, **HCl, HBr, HI** are hydracids.

10. (d) bluish-green

When **copper oxide (CuO)** reacts with **dilute hydrochloric acid (HCl)**, it forms **copper chloride (CuCl₂)**, which gives a **bluish-green solution**:

11. (a) Potassium hydroxide

Potassium hydroxide (KOH) is a **strong base** because it **dissociates completely** in water into K^+ and OH^- ions:



Bases like $\text{Mg}(\text{OH})_2$, $\text{Ca}(\text{OH})_2$, and NH_4OH are **partially soluble** or weak bases, so they do not dissociate completely.

12. (d) Because it is a deliquescent substance

Zinc chloride (ZnCl_2) is **deliquescent**, meaning it **absorbs moisture from the air** and can dissolve in the absorbed water. To prevent this and keep it dry, it is stored in **airtight bottles**.

13. (c) Carbon monoxide

- Nitrogen dioxide (NO_2) + $\text{H}_2\text{O} \rightarrow \text{HNO}_3 + \text{HNO}_2$ (acids formed)
- Carbon dioxide (CO_2) + $\text{H}_2\text{O} \rightarrow \text{H}_2\text{CO}_3$ (carbonic acid)
- Sulphur trioxide (SO_3) + $\text{H}_2\text{O} \rightarrow \text{H}_2\text{SO}_4$ (sulfuric acid)
- **Carbon monoxide (CO) does not react with water** to form an acid.

Thus, **CO** is the only one that does not produce an acid.

14. (a) Bases are sour in taste. 

- Bases are **bitter in taste** (commonly mentioned as a property in exams, but some references may say sour for certain basic salts), and they turn **red litmus blue**.
- pOH is indeed defined as $-\log[\text{OH}^-]$ but in this question, the intended correct property-based answer is (a).

15. hydrochloric acid

16. sulphuric acid

17. hydrochloric acid

18. dirty green

19. acidic

20. Normal salt

21. (i) hydronium

(ii) hydroxide

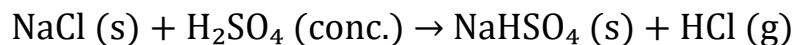
(iii) salt (iv) water

(v) hydrogen

22. (a) **Concentrated sulphuric acid** acts as a dehydrating agent.

(b) **Ammonia** reacts with chlorine to give **hydrogen chloride** and nitrogen.

23. (a) **Balanced chemical equation:**



(b) **Method of collection:**

Hydrogen chloride gas is collected by **downward delivery** (because it is denser than air).

(c) **Property of sulphuric acid:**

Concentrated sulphuric acid is **non-volatile** and has a **high boiling point**, which prevents it from escaping as vapor. It also acts as a **dehydrating agent**, making it suitable for this reaction.

24. Concentrated sulphuric acid.